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CYPHER 2026 · HACKATHON CHALLENGE PACK · 18 HOURS

Agentic AI for the Supply Chain

Eleven businesses. Eleven messy mornings.

Build an AI agent that notices the problem, weighs the options and gets the next move ready for a human to approve.

PRESENTED BY KD'S GARAGE
PARTICIPANT EDITION

READ THIS FIRST

Eleven businesses. Eleven messy mornings. One question each.

Every challenge in this pack starts with a note from someone inside a business: a buyer, a planner, a store head, a hospital superintendent. They are drowning in data and short on time. Your job is to build an AI agent that reads what they can't, works out what matters, weighs the options and gets the next action ready for them to approve.

The companies are fictional. The problems are real.

WHAT "AGENTIC" MEANS HERE

A chatbot that only summarises information will not score well. Your agent should run the full loop:

01	02	03	04	05	06
Observe	**Reason**	**Evaluate**	**Decide**	**Act**	**Explain**
Read and combine the data	Work out what is wrong and why	Compare the possible actions	Pick one, with numbers to back it	Draft the PO, transfer, alert or task	Show the evidence behind it

A human always gets the last word: every simulated action needs someone to approve it before it runs.

GROUND RULES

01 Use any language, LLM, agent framework, database or interface. APIs and local models are both fine.

02 A simple app is completely acceptable. Judges score decisions, not the size of your stack.

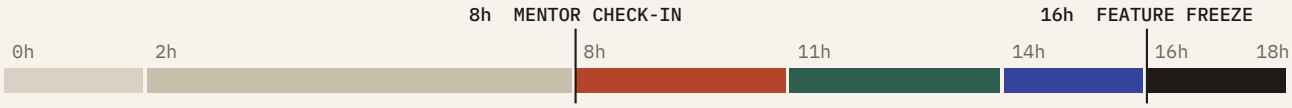
03 Design your agent around the records each business keeps, listed in your challenge. You may build your own sample data in that shape.

04 One agent or several specialised agents, your call.

05 Don't spend your 18 hours on login systems, Kubernetes, heavy infrastructure, complex ML pipelines or real integrations.

YOUR 18 HOURS

How the clock runs



Read & explore
0h to 2h
Read your challenge, map the problem, write one line: “Our agent decides ___.”

Build the core
2h to 8h
Set up your data, detect issues, agent logic, a basic screen

Make it decide
8h to 11h
Compare options, simulate actions, add the approval step

Make it explain
11h to 14h
Reasoning, more scenarios, edge cases

Test & freeze
14h to 16h
Run every scenario you can think of, freeze features, rehearse

Demo
16h to 18h
3 to 5 minutes per team

Expect the unexpected. Judges won’t only watch the demo you rehearsed. They will try your agent on situations you didn’t plan for. If it genuinely reasons, it will cope. If it was tuned to one scenario, it won’t.

HOW YOU’RE SCORED



100 POINTS

25	**Agentic behaviour**	Does it observe, reason, compare, decide and act, or does it only chat and display?
20	**Domain reasoning**	Does it understand the business problem in your challenge? (See each challenge.)
20	**Decision quality**	Are the recommendations practical, backed by numbers and commercially sensible?
20	**Working prototype**	Does the full flow run, including on situations the team didn’t plan for?
10	**Innovation**	A useful idea beyond what the challenge asks for
5	**Demo & clarity**	Is it easy to understand and follow?

A simple agent that makes the right call and shows its working beats a beautiful one that only talks. Aim for the moment a judge thinks: I would use this at work tomorrow.

THE CHALLENGES

Pick your morning

01	**The Spare Parts Desk** Kaveri Spares & Hydraulics · North Karnataka, 6 stores and 2 warehouses	*"What supply-chain problems need my attention today, and what should I do about them?"*
02	**The Festival Rush** VoltKart Electronics · Multi-store electronics retailer with a central warehouse	*"What needs my attention today, and what should I do about it?"*
03	**The Wrong Sizes** BigFit Apparel · Multi-store clothing retailer, sizes XL to 5XL	*"What inventory decisions should we make today to keep the right sizes available without piling up old stock?"*
04	**Head Office, 8:30 a.m.** FreshBasket Franchise Network · Fresh food and grocery franchise stores	*"Which stores need my attention today, what is causing the problem, and what should I do?"*
05	**The Blocked Bed** Sanjeevani Multispeciality Hospital · Multi-department hospital	*"What hospital operations need my attention today, and what should we do?"*
06	**Before the Match Starts** Zipcart · 10-minute grocery delivery, 30 dark stores in one city	*"Which stores will run out of what before the next truck, and what should I do right now?"*
07	**Batch B2231** Arogya Pharma Distributors · 3 warehouses serving 400 chemists and 20 hospitals	*"Which batches, customers and medicines need action today, and what should we do?"*
08	**The Truck Near Hubli** Sudha Home Products · 2 plants, 15 depots, 6 transport carriers	*"Which shipments are going wrong, which ones actually matter, and how do we recover?"*
09	**One Missing Part** Precision Axle Components · Auto-component plant, one assembly line	*"Which production runs are at risk from material shortages, which customer orders does that hit, and what should I do?"*
10	**What Came Back** Gadgetbay · E-commerce seller, returns centre	*"What should we do with today's returns, and what are they telling us about our products and suppliers?"*
11	**Three Small Warnings** Malnad Industrial Pumps · Manufacturer buying from 150 suppliers	*"Which suppliers are becoming a risk, why, and what should procurement do about it?"*

01 The Spare Parts Desk

Kaveri Spares & Hydraulics
North Karnataka, 6 stores and 2 warehouses

FROM Ramesh Kulkarni, Head of Purchasing

Every morning I open five spreadsheets: stock, sales, suppliers, open POs and whatever the stores have messaged me overnight. By the time I have lined them up, Gokak has already run out of something and Belgaum is sitting on a pile of filters nobody needs.

I don't need another report. I need someone who reads all of it before I do, tells me which three things actually matter today, and has a draft PO or transfer ready for me to approve.

"What supply-chain problems need my attention today, and what should I do about them?"

YOUR BRIEF

An agent that reads inventory, sales, supplier and purchase-order data, finds the problems, weighs the options and prepares the next action for a human to approve. You are not building an ERP. You are building the colleague Ramesh wishes he had.

WHAT THE BUSINESS KEEPS TRACK OF

Products	`sku, name, machine_model, category`
Inventory	`sku, location, stock`
Sales	`date, sku, location, qty_sold`
Suppliers	`supplier, sku, price, lead_time_days, moq`
Purchase orders	`po, supplier, sku, qty, expected_date, status`

PROBLEMS YOUR AGENT SHOULD SPOT

01 Parts that will run out before any supplier can deliver
02 Slow-moving stock tying up cash
03 Purchase orders that are overdue and putting stock at risk
04 Suppliers whose price, lead time or MOQ make them the wrong choice for an urgent need
05 Demand that has suddenly jumped or dropped at one location

A MORNING AT KAVERI SPARES & HYDRAULICS

Gokak has 8 units of a filter and has been selling 4 a day. The usual supplier needs 7 days. A second supplier can deliver in 3 days, but charges more. Belgaum also stocks the filter.

A STRONG AGENT

It works out that Gokak runs dry in about 2 days, long before the usual supplier can help. It checks Belgaum first. If a transfer won't work, it drafts a PO with the faster supplier and flags the extra cost for Ramesh to approve.

YOUR AGENT MUST

- ☐ Answers “what needs my attention today?” in a dashboard or chat
- ☐ Finds at least three different kinds of problem in the data
- ☐ Explains each one with numbers: demand rate, stock, lead time, price, open orders
- ☐ Compares at least two options before recommending one
- ☐ Produces at least one simulated action: a PO draft, transfer request, supplier enquiry or alert
- ☐ Lets a person approve or reject before anything is executed

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LEAVE OUT Login systems, ERP modules, ML forecasting pipelines, real supplier integrations.

DOMAIN POINTS (20) Judges look for an understanding of stock, demand, lead times, suppliers and the trade-offs between them.

02 The Festival Rush

VoltKart Electronics
Multi-store electronics retailer with a central warehouse

FROM Ananya Rao, Regional Supply Manager

We sell everything from earphones to air conditioners, across a dozen stores. Demand moves with weekends, festivals, launches, price drops and whatever the store next door is running.

Our system will happily tell me stock is low. What it won't tell me is that the same TV is gathering dust in another store, that one supplier is faster but costs more, and that a promotion starts on Friday. I need something that puts those pieces together.

"What needs my attention today, and what should I do about it?"

YOUR BRIEF

An agent that watches sales, stock, suppliers, purchase orders and promotions across stores, connects the facts, compares the options and recommends one, with a simulated action ready for approval.

WHAT THE BUSINESS KEEPS TRACK OF

Products	sku, product, brand, category, model, selling_price, launch_date
Inventory	sku, store, stock, ageing_days
Sales	date, sku, store, qty_sold, selling_price
Suppliers	supplier, sku, purchase_price, lead_time_days, moq, availability
Purchase orders	po, supplier, sku, qty, expected_date, status
Promotions	sku_or_category, start, end, discount, expected_uplift

PROBLEMS YOUR AGENT SHOULD SPOT

01 Fast sellers that will run out before replenishment
02 Ageing stock that is losing value every week
03 Demand spikes from promotions, weekends or events
04 Older models about to lose demand to a new launch
05 Supplier choices that trade price against lead time and MOQ
06 One store overstocked while another is nearly out
07 Late purchase orders that create a stock gap

A MORNING AT VOLTKART ELECTRONICS

A 55-inch smart TV: Store A has 4 units and sells 2 a day. A festival promotion starts in 3 days. Supplier A takes 7 days. Supplier B takes 2 days and costs 5% more. Store B has 12 units and sells half a unit a day.

A STRONG AGENT

It sees the stock-out and the promotion coming together, checks Store B before buying anything, works out whether a transfer covers the promotion, and compares that against Supplier B. Then it recommends one plan and says why.

YOUR AGENT MUST

- ☐ Answers “what needs my attention today?” in a dashboard or chat
- ☐ Finds at least three different kinds of problem in the data
- ☐ Shows its evidence: sales velocity, days of stock, lead time, ageing, promotion impact
- ☐ Compares at least two options when alternatives exist
- ☐ Produces at least one simulated action: PO draft, inter-store transfer, supplier enquiry or alert
- ☐ Lets a person approve or reject before anything is executed

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LEAVE OUT A retail ERP, login systems, complex ML pipelines, real supplier integrations.

DOMAIN POINTS (20) Judges look for an understanding of inventory, demand, ageing, promotions, suppliers and store balancing.

03 The Wrong Sizes

BigFit Apparel
Multi-store clothing retailer, sizes XL to 5XL

FROM Farhan Sheikh, Merchandising Lead

Our stores can show 20 shirts on the shelf and still lose the customer, because 17 of them are the wrong size. Our customers mostly wear XL and up, and those sizes sell out first.

Meanwhile another store has a stack of XXL nobody is buying. Add festivals, weather and styles that go stale in a season, and it becomes very hard to know whether to transfer, reorder or mark down.

"What inventory decisions should we make today to keep the right sizes available without piling up old stock?"

YOUR BRIEF

A size-aware agent that looks at stock by style, colour, size and store, spots where the mix is wrong, and recommends a transfer, reorder, markdown or wait, with a simulated action ready for approval.

WHAT THE BUSINESS KEEPS TRACK OF

Products	sku, product, category, style, colour, size, season, selling_price
Inventory	sku, store, stock, ageing_days
Sales	date, sku, store, qty_sold, selling_price
Stores	store, location, store_type, customer_profile
Suppliers	supplier, sku, purchase_price, lead_time_days, moq
Purchase orders	po, supplier, sku, qty, expected_date, status
Promotions	product_or_category, start, end, discount, expected_uplift

PROBLEMS YOUR AGENT SHOULD SPOT

01 Large sizes in high demand and running low
02 Sizes or styles overstocked in one store and short in another
03 Old styles or colours that need a markdown or transfer
04 Sudden rises or falls in demand for a size
05 Festivals or promotions that will need extra stock
06 Supplier choices across price, lead time and MOQ
07 Styles and size combinations to reorder, cut back or drop

A MORNING AT BIGFIT APPAREL

Style S101, a men’s casual shirt: Store A has 18 units, but only 2 in XXL and 1 in 3XL. Those sizes sell three times faster than M and L. Store B has 12 XXL and sells them slowly. A festival promotion starts in four days.

A STRONG AGENT

It sees past the healthy-looking total to the broken size mix. It moves XXL from Store B to Store A, estimates the promotion lift, and only suggests a reorder if the whole network can’t cover it.

YOUR AGENT MUST

- ☐ Answers “what needs my attention today?” in a dashboard or chat
- ☐ Finds at least three different kinds of problem in the data
- ☐ Looks at stock by size, never only by total
- ☐ Shows its evidence: size demand, sales velocity, ageing, promotion impact, store imbalance
- ☐ Compares transfer, reorder, markdown, promotion and wait
- ☐ Produces at least one simulated action and lets a person approve it first

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LEAVE OUT An ERP, billing, login systems, real supplier integrations.

DOMAIN POINTS (20) Judges look for an understanding of size mix, demand, ageing, seasonality, stores and replenishment.

04 Head Office, 8:30 a.m.

FreshBasket Franchise Network
Fresh food and grocery franchise stores

FROM Kavitha Menon, Head of Franchise Operations

I get reports from every store: sales, stock, wastage, staff, footfall, inspections. What I don't get is an answer to the only question that matters at 8:30 in the morning: which stores need me today, and why?

A store can post decent sales while quietly throwing away a fortune in fresh food. Another has plenty of stock but customers keep finding empty shelves. The numbers are all there. Nobody is joining them up.

"Which stores need my attention today, what is causing the problem, and what should I do?"

YOUR BRIEF

An agent that reads every store's signals, ranks stores by urgency, investigates the likely cause by connecting the signals, and recommends an action. Head Office gets a ranked list; each store manager gets a short to-do list for the day.

WHAT THE BUSINESS KEEPS TRACK OF

Stores	store, location, franchisee, format, status
Products	sku, category, price, perishability, shelf_life_days
Sales	date, store, sku, qty_sold, revenue
Inventory	store, sku, stock, reorder_level
Wastage	store, sku, qty_wasted, reason, date
Purchase orders	po, supplier, sku, qty, expected_date, status
Customers	store, footfall, transactions, avg_bill_value
Staff	store, attendance, staffing_level, productivity
Compliance	store, checklist_score, open_issues, last_inspection

PROBLEMS YOUR AGENT SHOULD SPOT

01 Stores with unusual drops in sales, margin, transactions or bill value
02 Fast movers about to run out
03 Wastage that is unusual or climbing
04 Stock likely to expire before it sells
05 What to order, how much and when
06 Stores with repeated SOP or compliance failures
07 Staffing or attendance problems hurting performance
08 Footfall and complaints that explain a performance drop

A MORNING AT FRESHBASKET FRANCHISE NETWORK

Store 17: sales are down 18%. Footfall is down only 5%, but transactions are down 15%. Fresh-food wastage is up 28%, 12 fast movers are short on the shelf, and a supplier delivery is late.

A STRONG AGENT

It doesn't just say "sales are down". It links the late delivery and empty shelves to lost transactions, and notices that ordering habits are overstocking other fresh categories. It recommends fixing both, in order of impact.

YOUR AGENT MUST

- ☐ A Head Office view ranking stores by urgency
- ☐ A store view answering "what do I need to do today?"
- ☐ Finds at least three different kinds of problem
- ☐ Explains causes by connecting signals, not by listing KPIs
- ☐ Recommends actions: replenishment, markdown, wastage fix, task or escalation
- ☐ Produces at least one simulated action and lets a manager approve it first

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LEAVE OUT A POS, ERP, HR system, login systems, real supplier integrations.
DOMAIN POINTS (20) Judges look for an understanding of store performance, inventory, wastage, customers, staff and compliance.

05 The Blocked Bed

Sanjeevani Multispeciality Hospital
Multi-department hospital

FROM Dr. Suresh Iyer, Medical Superintendent

We have screens for everything: beds, appointments, labs, discharges, rosters, complaints. Each one tells me something is late. None of them tell me why the emergency department is backed up today.

Sometimes we have beds free, just not in the ward we need. A doctor has a waiting room full of patients while another department is quiet. A discharge is held up because one report hasn't come back. I want the bigger picture, fast.

"What hospital operations need my attention today, and what should we do?"

YOUR BRIEF

An operations agent that connects signals across departments, finds the real bottleneck, ranks issues by urgency and recommends actions such as rescheduling, escalation, staff reallocation or task priority, all reviewed by authorised staff.

WHAT THE BUSINESS KEEPS TRACK OF

Patients	patient_id, age_group, department, visit_type, status
Appointments	doctor, department, datetime, status, waiting_time
Beds	ward, bed_type, occupancy, availability, admission_discharge_status
Staff	department, shift, availability, workload
Diagnostics	test, patient, status, requested_time, expected_completion
Admissions	admission_time, department, ward, expected_discharge
Discharge	request, pending_tasks, medication_status, document_status
Feedback	complaint, category, severity, resolution_status

PROBLEMS YOUR AGENT SHOULD SPOT

01 Departments or time slots with unusually long waits
02 Bed shortages, idle beds, or beds in the wrong ward
03 Overloaded doctors next to spare capacity elsewhere
04 Tests and reports holding up patient movement
05 Discharges stuck on pending tasks
06 Cancellations, no-shows and overloaded slots
07 Complaints that trace back to an operational problem
08 Situations that need escalating to the right team

A MORNING AT SANJEEVANI MULTISPECIALITY HOSPITAL

The emergency department's wait time has jumped. Five beds are occupied, two are blocked by discharges that are waiting on paperwork, several lab reports are late, and the next shift is short-staffed.

A STRONG AGENT

It sees that the problem isn't simply too many patients. Stuck discharges and late reports are eating bed capacity. It recommends clearing the pending discharge tasks, escalating the late reports and warning the shift manager about staffing.

YOUR AGENT MUST

- ☐ A dashboard highlighting which departments need attention
- ☐ Finds at least three different kinds of problem
- ☐ Explains causes by connecting signals, not by listing KPIs
- ☐ Ranks issues by urgency and impact
- ☐ Recommends actions and produces at least one simulated task, alert or priority queue
- ☐ Keeps every operational or clinical action reviewable by authorised staff

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LEAVE OUT A hospital information system, billing, login systems, real clinical integrations. The agent is never presented as a doctor: diagnosis, treatment and medication stay with qualified people.

DOMAIN POINTS (20) Judges look for an understanding of patient flow, beds, appointments, diagnostics, staff and discharges.

06 Before the Match Starts

Zipcart
10-minute grocery delivery, 30 dark stores in one city

FROM Nikhil Bhat, City Operations Manager

Each of our dark stores carries about 3,000 products in a space the size of a large flat. Trucks come from the main warehouse a few times a day. Between trucks, whatever is on the shelf is all we have.

Demand turns on a dime. A cricket match, a rainy evening, a festival, and a category triples in an hour. And there is no spare shelf. If I want more cold drinks in, something else has to come out.

"Which stores will run out of what before the next truck, and what should I do right now?"

YOUR BRIEF

An agent that projects stock hour by hour, factors in events, respects shelf space, and recommends a transfer, an earlier truck, a shelf swap or waiting, with a simulated instruction ready for approval.

WHAT THE BUSINESS KEEPS TRACK OF

Products	sku, product, category, size, selling_price, margin
Store inventory	store, sku, stock, shelf_slot, slot_capacity
Orders	timestamp, store, sku, qty (hourly)
Warehouse stock	sku, stock
Truck schedule	store, slot_time, truck_capacity
Riders	store, hour, riders_available, orders_per_rider_hour
Events	date, time, event_type, expected_uplift_by_category

PROBLEMS YOUR AGENT SHOULD SPOT

01 Products that will run out before the next truck
02 Matches, festivals or rain about to lift demand
03 Slow sellers sitting in shelf space that fast sellers need
04 A nearby store with surplus of something another store is about to lose
05 Stores losing orders because of too few riders, not too little stock

A MORNING AT ZIPCART

Store 7 has 14 bottles of a 750 ml cold drink and sells 9 an hour on match nights. The match starts at 7:30 pm. The next truck comes at 9 pm. Store 9, 2 km away, has 60 and sells 3 an hour. Store 7’s drink shelf is full, and the slot next to it holds a snack that sold 4 packs all week.

A STRONG AGENT

It works out the store runs dry well before 9 pm once the match begins. It compares pulling the truck forward, a transfer from Store 9 and doing nothing, puts a number on lost sales for each, and suggests clearing the slow snack to make room.

YOUR AGENT MUST

- ☐ Answers “what needs my attention in the next 4 hours?”
- ☐ Finds at least three different kinds of problem
- ☐ Works with hourly demand, not daily averages
- ☐ Never recommends more stock than the shelf can hold
- ☐ Compares at least two options: transfer, earlier truck, shelf swap or wait
- ☐ Produces at least one simulated instruction and lets a manager approve it first

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LEAVE OUT A delivery app, rider routing, a customer storefront, live integrations. Hourly records are enough.
DOMAIN POINTS (20) Judges look for an understanding of hourly demand, event impact, shelf-space trade-offs, store balancing, and telling a stock problem from a rider problem.

07 Batch B2231

Arogya Pharma Distributors
3 warehouses serving 400 chemists and 20 hospitals

FROM Meera Joshi, Quality & Compliance Head

In our business, “we have 500 strips” tells me almost nothing. I need to know which batches, expiring when, stored at what temperature, and shipped to whom.

Manufacturers take back near-expiry stock only within a window. Cold rooms occasionally drift above 8°C overnight. And a recall can land on my desk at any hour. When it does, I need to know within minutes exactly where every unit went.

“Which batches, customers and medicines need action today, and what should we do?”

YOUR BRIEF

An agent that tracks stock at batch level, traces any batch from warehouse to customer, links temperature breaches to the batches affected, and recommends what to block, return, move or reorder, with every decision approved by a qualified person.

WHAT THE BUSINESS KEEPS TRACK OF

Products	sku, molecule, brand, category, storage (ambient / 2-8C), critical_drug
Batch inventory	sku, batch, warehouse, qty, mfg_date, expiry_date
Dispatches	date, customer, sku, batch, qty
Customers	customer, type (chemist / hospital), location, credit_terms
Temperature logs	warehouse, cold_room, timestamp, temp_c
Suppliers	manufacturer, sku, lead_time_days, moq, return_window_days
Purchase orders	po, manufacturer, sku, qty, expected_date, status
Recalls	date, sku, batches, reason, recall_class

PROBLEMS YOUR AGENT SHOULD SPOT

01 Batches that will expire before they sell
02 Near-expiry stock whose return window is about to close
03 Cold-room breaches and the batches stored there at the time
04 Customers who received a recalled batch, and how much is still in the warehouse
05 Critical medicines likely to run short
06 Newer batches being dispatched while older ones sit unsold

A MORNING AT AROGYA PHARMA DISTRIBUTORS

A recall notice arrives for Batch B2231 of an antibiotic. The warehouse still has 180 units. Over the last 30 days, 640 units went to 23 chemists and 2 hospitals. A clean batch, B2240, has 400 units.

A STRONG AGENT

It blocks B2231 from further dispatch, lists every affected customer and quantity, and drafts the notices. It checks whether B2240 can cover replacements plus normal demand. If not, it drafts an urgent PO and suggests serving hospitals first.

YOUR AGENT MUST

- ☐ Answers “what needs my attention today?”
- ☐ Tracks stock by batch and expiry, not only by total
- ☐ Finds at least three different kinds of problem
- ☐ Traces any batch: where it came from, where it is, who received it
- ☐ Compares at least two options: discount, return, transfer, urgent PO or allocation
- ☐ Produces at least one simulated action, approved by a pharmacist or compliance officer first

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LEAVE OUT Billing, a drug database, real regulatory integrations. The agent never decides on its own whether a medicine is safe to use; it gathers the evidence and hands the call to a qualified person.

DOMAIN POINTS (20) Judges look for an understanding of batches, expiry, cold chain, recalls, dispatch order, critical-drug priority and return windows.

08 The Truck Near Hubli

Sudha Home Products
2 plants, 15 depots, 6 transport carriers

FROM Prakash Hegde, Logistics Manager

On any given day, a few of our trucks are running late. A breakdown on the highway, a carrier that didn't turn up, a load stuck at a checkpost. Most delays don't matter. A few matter a lot.

What I can't see quickly is which ones. A truck that is 18 hours late to a depot with a week of stock is fine. The same delay to a depot with half a day of stock and a big retailer order tomorrow is a crisis.

"Which shipments are going wrong, which ones actually matter, and how do we recover?"

YOUR BRIEF

A control-tower agent that links every delay to depot stock and customer commitments, ranks problems by business impact, and recommends the cheapest recovery that protects priority customers.

WHAT THE BUSINESS KEEPS TRACK OF

Shipments	shipment_id, origin, depot, carrier, skus, qty, dispatch_time, planned_arrival
Shipment status	shipment_id, timestamp, status, revised_eta
Carriers	carrier, lanes, rate, truck_types, on_time_history
Depot inventory	depot, sku, stock, avg_daily_demand
Plant stock	plant, sku, stock
Commitments	customer, depot, sku, qty, promised_date, priority
Lanes	origin, destination, distance_km, transit_hours

Shipment status is recorded as status updates. No GPS or maps needed.

PROBLEMS YOUR AGENT SHOULD SPOT

01 Delays that land after a depot runs out
02 Customer orders that will be late because of a delay
03 Carriers whose on-time record is slipping
04 Half-empty trucks to nearby depots that could be combined
05 Depots or plants that could cover a gap

A MORNING AT SUDHA HOME PRODUCTS

A truck with 1,200 cases of detergent for the Hubli depot has broken down and will arrive 18 hours late. Hubli has 300 cases and sells 25 an hour. Dharwad depot, 20 km away, has 900 and sells 10 an hour. A key retailer expects 400 cases from Hubli tomorrow.

A STRONG AGENT

It spots a 6-hour stock-out gap and a retailer order at risk. It compares a transfer from Dharwad, an emergency truck from the plant with another carrier, and part-filling the retailer, puts cost against revenue at risk, and recommends one plan with a fallback.

YOUR AGENT MUST

- ☐ A control-tower view or chat answering “what needs my attention today?”
- ☐ Finds at least three different kinds of problem
- ☐ Links every delay to depot stock and customer orders
- ☐ Ranks issues by business impact, not by hours late
- ☐ Compares at least two recovery options with cost and effect
- ☐ Produces at least one simulated action and lets a manager approve it first

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LEAVE OUT Route optimisation, GPS, maps, a transport management system, real carrier integrations.
DOMAIN POINTS (20) Judges look for an understanding of transit times, delay impact, depot stock cover, carrier trade-offs and customer priority.

09 One Missing Part

Precision Axle Components
Auto-component plant, one assembly line

FROM Vinay Patil, Production Planning Head

We make five products on one line. Each needs 10 to 20 components. If 19 are on the rack and one isn't, the line stops. It doesn't matter how much of everything else we have.

Every morning I look two weeks ahead and try to spot the part that will stop us. Usually I find it the day before, when the only options left are expensive ones.

"Which production runs are at risk from material shortages, which customer orders does that hit, and what should I do?"

YOUR BRIEF

An agent that explodes the schedule through the bill of materials, finds the first shortage, links it to the customer orders affected, and recommends the cheapest way to keep promises: a partial run, a schedule swap, a substitute or an expedited PO.

WHAT THE BUSINESS KEEPS TRACK OF

Products	product, description, selling_price, margin
Bom	product, component, qty_per_unit (single level)
Schedule	run_id, date, product, qty, line
Components	component, stock, unit_cost
Substitutes	component, substitute, notes, extra_cost
Suppliers	supplier, component, price, lead_time_days, moq, expedite_cost
Purchase orders	po, supplier, component, qty, expected_date, status
Customer orders	order, customer, product, qty, due_date, priority, late_penalty

PROBLEMS YOUR AGENT SHOULD SPOT

01 Runs that need more of a component than will be in stock that day
02 Late POs that will stop a specific run
03 Customer orders that will miss their due date
04 Approved substitutes in stock that could fill a gap
05 Components tied up for products nobody is ordering

A MORNING AT PRECISION AXLE COMPONENTS

Tuesday's run needs 500 of component C-114, but only 320 are in stock. The PO for 1,000 is 4 days late. Substitute C-114A has 150 in stock at 8% more cost. Thursday's run, a different product, has all its parts. Tuesday's output goes to a priority customer with a late penalty.

A STRONG AGENT

It finds a 180-unit gap and sees the substitute only covers 150. It compares a partial run, swapping Tuesday and Thursday, using the substitute, expediting the PO and combinations, shows which orders each option affects, and recommends one, such as swapping runs and expediting.

YOUR AGENT MUST

- ☐ Answers "what is at risk in the next 2 weeks?"
- ☐ Calculates component needs from schedule and BOM, not just stock levels
- ☐ Finds at least three different kinds of problem
- ☐ Links every shortage to the customer orders it affects
- ☐ Compares at least two options: partial run, swap, substitute or expedite
- ☐ Produces at least one simulated action and lets the planner approve it first

LEAVE OUT A full MRP system, multi-level BOMs, shop-floor control, capacity scheduling. One line, one BOM level and two weeks are enough.

DOMAIN POINTS (20) Judges look for an understanding of BOM requirements, lead times, substitutes, schedule flexibility and the cost of late orders.

10 What Came Back

Gadgetbay
E-commerce seller, returns centre

FROM Sneha Reddy, Returns & Recovery Lead

About 8 in every 100 orders come back to us. Each return can be restocked, refurbished, sent back to the vendor, sold to a liquidator or scrapped. Each route gets us a different amount of money, at a different speed.

Right now most returns sit on racks until someone gets to them, and they lose value every week they wait. And when 30 people return the same headphones for the same fault, that should be telling us something about the supplier.

"What should we do with today's returns, and what are they telling us about our products and suppliers?"

YOUR BRIEF

An agent that works out the net recovery and time-to-cash of every route for each return, picks the best mix (including splitting a batch), and spots return patterns that point to a supplier quality problem.

WHAT THE BUSINESS KEEPS TRACK OF

Products	sku, category, cost_price, selling_price, supplier
Returns	return_id, date, sku, supplier_batch, reason_code, grade (A/B/C/D)
Sales	date, sku, qty_sold, selling_price
Inventory	sku, new_stock, avg_daily_demand
Refurbishment	category, cost, days_required, resale_pct_of_new
Vendor terms	supplier, sku, rtv_allowed, credit_pct, window_days
Liquidation	category, grade, recovery_pct_of_cost

PROBLEMS YOUR AGENT SHOULD SPOT

01 High-value returns waiting too long
02 Items headed for liquidation that would earn more restocked or refurbished
03 Vendor return windows about to close
04 SKUs or supplier batches with unusually high returns
05 Refurbishing what nobody wants, or liquidating what people do want

A MORNING AT GADGETBAY

40 wireless headphones came back, graded B. New ones sell for ₹2,500 and cost ₹1,400. Refurbishing costs ₹250 each, takes 5 days and resells at 75% of new. The vendor takes returns within 30 days for 60% credit; 24 units are still in that window. Liquidators pay 35% of cost. New stock already covers 60 days of demand. 70% of these returns report the same fault, all from batch HB-09.

A STRONG AGENT

It calculates what each route really recovers, notes that refurbished units will compete with two months of new stock, and splits the batch: the 24 eligible units back to the vendor, the rest refurbished. It also raises a quality alert on HB-09.

YOUR AGENT MUST

- ☐ Answers “what needs my attention today?”
- ☐ Finds at least three different kinds of problem
- ☐ Calculates net recovery and time-to-cash per route
- ☐ Connects return reasons to products or supplier batches
- ☐ Compares at least two routes and allows a batch to be split
- ☐ Produces at least one simulated action and lets a manager approve it first

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LEAVE OUT A customer returns portal, refunds, warehouse management, marketplace integrations. Condition grades are recorded at intake, so no image inspection needed.

DOMAIN POINTS (20) Judges look for an understanding of recovery value, time-to-cash, vendor terms, demand fit, and returns as a quality signal.

11 Three Small Warnings

Malnad Industrial Pumps
Manufacturer buying from 150 suppliers

FROM Deepak Shetty, Procurement Manager

My supplier problems never arrive as one big alarm. A price creeps 7% above contract. A couple of lots get rejected. A certificate is about to expire. Each lives in a different system, and on its own each looks manageable.

Then one day they all turn out to be the same supplier, for a part only they make, and we have three weeks of stock left.

"Which suppliers are becoming a risk, why, and what should procurement do about it?"

YOUR BRIEF

An agent that combines price, quality, delivery and compliance signals for every supplier, weighs them against how critical the item is and how much stock is left, and recommends whether to continue, hold, switch or bring in a second source.

WHAT THE BUSINESS KEEPS TRACK OF

Suppliers	supplier, category, location, approval_status
Items	item, criticality (A/B/C), approved_suppliers
Contracts	supplier, item, contract_price, valid_from, valid_to
Purchase orders	po, supplier, item, qty, price_paid, order_date, promised_date, received_date
Inspections	supplier, item, lot, date, qty_inspected, qty_rejected, defect_type
Compliance docs	supplier, doc_type, number, expiry_date
Item demand	item, monthly_use, stock

PROBLEMS YOUR AGENT SHOULD SPOT

01 Prices paid above contract, or rising faster than similar items
02 Rejection rates climbing over recent lots
03 On-time delivery slipping or lead times stretching
04 Certificates and registrations expiring on active suppliers
05 Critical items with only one approved supplier
06 Several of these pointing at the same supplier

A MORNING AT MALNAD INDUSTRIAL PUMPS

The only approved supplier of a critical casting has an ISO certificate expiring in 10 days. Its last two lots were rejected at 6% and 9%, against a usual 1%. The last three POs were paid 7% over contract. Stock covers 25 days. Two other foundries make similar castings but aren't approved for this part yet.

A STRONG AGENT

It sees that none of the warnings is serious alone, but together, on a part with no backup, they are. It compares tighter inspection, a compliance hold and starting a second source, drafts an RFQ to the two foundries, and checks whether approval can happen inside 25 days.

YOUR AGENT MUST

- ☐ Answers "which suppliers need my attention today?"
- ☐ Finds at least three different kinds of problem
- ☐ Combines price, quality, delivery and compliance into one view per supplier
- ☐ Ranks risk by item criticality and stock cover, not supplier score alone
- ☐ Compares at least two options: continue, hold, switch, dual-source or renegotiate
- ☐ Produces at least one simulated action and lets the manager approve it first

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LEAVE OUT A supplier portal, e-auctions, contract management, real document checks.

DOMAIN POINTS (20) Judges look for an understanding of contract pricing, supplier quality and delivery, compliance, single-source risk and the time it takes to approve a new supplier.

BEFORE HOUR 0

Pick your morning. Then plan your loop.

Choose the challenge whose business owner you understand best. Sketch how your agent will observe, reason, compare, decide and act. Decide who on your team builds what. Eighteen hours goes faster than you think.

Real businesses are messy. Some problems are urgent, some only look urgent, and the best agents can tell the difference.

See you at hour 0.

Cypher - Problem Statements

Rules for All Three Problems

UI is the most important criterion

A working feature behind a poor interface scores low. Every problem is judged first on how it looks, feels, and behaves on real devices.

- **Responsive by default:** every screen must work cleanly on mobile (360px), tablet (768px), and desktop (1280px+). No horizontal scrolling, no overlapping elements, no tiny tap targets.
- **Consistent design system:** one color palette, one type scale, reusable components, and clear spacing.
- **Complete states:** loading, empty, error, and success states for every screen.
- **Usability:** a new user should finish the main flow without instructions.
- **Accessibility basics:** readable contrast, keyboard navigation, labels on inputs, alt text.
- **Performance:** fast first load, optimized images, smooth interactions.

Extra bonus: Deployment

Teams that deploy get extra marks on top of their score.

Bonus item	What earns it
Live public URL	The app runs on a real hosted link (Vercel, Netlify, Render, Railway, AWS, GCP, or similar)

Production database	Hosted database, not a local file
Environment safety	Secrets in environment variables, never in the repo
CI/CD	Auto-deploy on every push to the main branch
Docker	A working Dockerfile or docker-compose setup
Domain and HTTPS	Valid SSL on a proper domain or subdomain
Monitoring	Basic logging, error tracking, or a health-check endpoint

Scoring split (same for all problems)

Area	Weight
UI, UX, and responsiveness	30%
Feature completeness (the 15 checkpoints)	30%

Innovation and uniqueness	15%
Code quality, architecture, and security	15%
Presentation and demo	10%
Deployment bonus	**up to +10% extra**

Submission checklist

1. Live URL (or a recorded demo if not deployed)
2. Source code repository with a clear README
3. Architecture diagram
4. A 3-minute demo video
5. Test logins for every role

Problem 1: Launch-Your-Store E-Commerce Builder

Fill a form, pick a theme, get a live online store.

The pitch

Build a platform where any small business owner can create a complete, live online store in minutes without writing code.

The flow

1. The owner signs up and enters **store name, logo, contact details, and basic business info**.
2. The owner chooses **categories** for the store.
3. The owner chooses how to add products: **import dummy products** for the chosen categories, or **upload an Excel/CSV sheet**.
4. The owner picks a **theme and UI style**.
5. The platform generates a **unique live URL** (for example, `storename.yourplatform.com`) where the store is running.
6. The owner manages everything from an **admin panel**.
7. An **AI chatbot** answers the owner's questions about their store data.

Why it is harder than it looks

- Multi-tenancy: hundreds of stores on one platform, each with isolated data.
- Excel files are messy: wrong headers, missing columns, bad prices, duplicate rows.
- The theme must change the whole storefront without rebuilding it.
- The chatbot must query real data accurately and never invent numbers.

15 Checkpoints

1. **Onboarding form:** collect store name, logo upload, contact details, address, and business type with validation and a clean step-by-step wizard.
2. **Category selection:** the owner picks from a predefined category list and can add custom categories.
3. **Dummy product import:** one click fills the store with realistic sample products (names, images, prices, descriptions) for the chosen categories.

4. **Excel/CSV upload:** import products from a sheet with a downloadable template, column mapping, preview, and a clear error report for bad rows.
5. **Theme selection:** at least 4 distinct themes with live preview before the owner confirms; colors, fonts, and layout change together.
6. **Unique live URL:** each store gets its own subdomain or path that is live immediately after setup.
7. **Complete storefront:** home page, category pages, product page, search, filters, cart, and checkout flow for shoppers.
8. **Admin panel dashboard:** sales summary, order count, revenue, low-stock alerts, and recent activity at a glance.
9. **Full content control:** the owner can edit logo, banners, colors, fonts, homepage sections, footer, and contact info from the admin panel with no code.
10. **Product and inventory management:** add, edit, delete, bulk edit, variants, stock, pricing, discounts, and images.
11. **Order management:** view orders, change status (placed, packed, shipped, delivered, cancelled), and send status updates to customers.
12. **Multi-tenant isolation:** one store's owner can never see or change another store's data; role-based access (owner, staff).
13. **AI chatbot for the owner who created stores:** answers questions in plain language, such as "What are my top 5 selling products this month?", "Which products are low on stock?", or "Revenue last week vs this week?", by querying the real database safely.
14. **Chatbot safety and honesty:** read-only queries, no access to other stores' data, shows the data behind each answer, and says "I don't have that data" instead of guessing.
15. **Fully responsive UI:** the onboarding wizard, the storefront, and the admin panel all work smoothly on mobile, tablet, and desktop.

Problem 2: Placement Training LMS for College Students

One platform that takes a student from resume to offer letter.

The pitch

Build a Learning Management System dedicated to placement preparation. Students practice, get evaluated by AI, and track their readiness. Colleges and trainers monitor progress at scale.

Users

- **Student:** learns, practices, and tracks readiness
- **Trainer or faculty:** assigns tests, reviews performance, gives feedback
- **Placement officer or admin:** manages batches, content, and reports

Why it is harder than it looks

- Resume analysis must give specific, useful advice, not generic tips.
- Interviews must adapt to the candidate's answers.
- A code editor must run code safely and judge it correctly.
- Analytics must turn raw attempts into meaningful readiness insights.

15 Checkpoints

1. **Authentication and roles:** login for students, trainers, and admins with separate dashboards.

2. **Student dashboard:** progress, upcoming tests, streaks, readiness score, and recommended next actions.
3. **Resume builder:** guided form with multiple professional templates, live preview, and PDF download.
4. **Resume analyzer:** upload a resume, get an ATS-style score, section-wise feedback, missing keywords, and role-specific suggestions, optionally against a pasted job description.
5. **AI mock interviewer:** a live chat or voice interview for HR, technical, and behavioral rounds that asks adaptive follow-up questions.
6. **Interview feedback report:** per-skill scores, strengths, weaknesses, and answer-level comments after each interview.
7. **MCQ test engine:** timed tests across aptitude, verbal, reasoning, and technical topics, with randomized questions and a clear test interface.
8. **MCQ analyzer:** topic-wise accuracy, time per question, weak-area detection, and a comparison against the batch average.
9. **Online code editor:** multi-language editor (at least Python, Java, C++, JavaScript) with run, custom input, and output panels.
10. **Coding practice with auto-judging:** problems with hidden test cases, pass/fail results, runtime, and difficulty levels.
11. **Learning modules:** structured courses on aptitude, DSA, core CS subjects, and soft skills, with video or reading content and completion tracking.
12. **Company-wise preparation:** question sets and roadmaps tagged by company (for example, service-based vs product-based).
13. **Leaderboard and gamification:** badges, streaks, points, and batch rankings to keep students engaged.
14. **Trainer and admin panel:** create tests and questions (including bulk upload), assign them to batches, track every student, and export reports.
15. **Fully responsive UI:** every module works well on mobile, tablet, and desktop, including the test screen and the code editor.

Add as many as you can (extra features)

- Group discussion practice with AI participants
- Communication and English fluency analyzer
- Personalized study plan based on weak areas
- Aptitude shortcuts and formula revision cards
- Mock placement drives with multi-round simulation
- Job and internship board with application tracking
- Anti-cheating checks for tests (tab-switch detection, camera checks)
- Certificates on course completion
- Alumni mentor connect and discussion forum
- LinkedIn profile and GitHub portfolio reviewer
- Offline-friendly study material downloads
- Multilingual support (English and Hindi)

Problem 3: Next-Generation Quick Commerce (10-Minute Delivery)

Build the app that quick commerce is still missing.

The pitch

Several companies already deliver groceries and essentials in 10 minutes. Build a quick-commerce platform that matches the speed but does something **unique** that existing apps do not do well, across all three sides: the **customer**, the **rider**, and the **admin**.

Users

- **Customer app:** browse, order, and track

- **Rider app:** accept deliveries, navigate, and deliver
- **Admin panel:** run operations, inventory, and analytics

Mandatory requirements

- **Notifications** across all three roles (push, in-app, or both)
- **Google Maps** integration (location, live tracking, routes)

Where teams can be unique (pick at least one clear differentiator)

- Smart substitutions when an item is out of stock
- Honest delivery-time prediction with live updates instead of a fixed "10 minutes"
- Order batching so riders carry multiple nearby orders efficiently
- Rider wellbeing: fair earnings view, safe routing, weather and traffic alerts
- Reduced packaging and a sustainability score
- Shared group carts for families, hostels, and offices
- Recipe-to-cart: pick a dish and get all ingredients in the cart
- Voice and regional-language ordering
- Dark-store stock prediction to cut waste and out-of-stock events
- Transparent pricing with no hidden fees

Why it is harder than it looks

- Live tracking, rider assignment, and notifications must stay in sync in real time.
- Delivery time depends on distance, stock, rider availability, and traffic.
- Three different interfaces must feel like one product.

15 Checkpoints

1. **Customer onboarding:** sign up and login with OTP or email, and a saved address using Google Maps location picker and autocomplete.

2. **Product catalogue:** categories, search, filters, product pages, and fast browsing built for quick decisions.
3. **Cart and checkout:** smooth cart, coupon support, delivery slot or instant option, and clear price breakdown.
4. **Serviceability check:** detect whether the customer's location is within a dark store's delivery radius using Google Maps.
5. **Order placement and status flow:** placed, accepted, packed, picked up, on the way, delivered, with real-time status updates.
6. **Live order tracking:** the customer sees the rider's live location on Google Maps with an estimated arrival time.
7. **Rider app onboarding and availability:** the rider logs in, goes online or offline, and sees their current status.
8. **Rider order assignment:** new orders are assigned or offered to the nearest suitable rider with accept and reject options.
9. **Rider navigation:** pickup and drop routes on Google Maps, with distance and time shown.
10. **Rider earnings and history:** per-delivery earnings, daily and weekly summary, and completed order history.
11. **Admin dashboard:** live orders, active riders on a map, delivery times, revenue, and cancellations.
12. **Inventory and catalogue management:** the admin manages dark stores, stock levels, pricing, offers, and low-stock alerts.
13. **Notifications for all roles:** the customer gets order updates, the rider gets new-order and payment alerts, and the admin gets low-stock and delay alerts.
14. **The unique feature:** at least one clearly demonstrated differentiator from the list above (or your own idea), explained with the problem it solves.
15. **Fully responsive UI:** the customer app is mobile-first and polished, the rider app is one-hand friendly, and the admin panel works on tablet and desktop.

Bonus ideas

- Payment in test mode with refunds on cancellation
- Ratings for products and riders
- Delay compensation or auto-refund rules
- Heatmap of demand in the admin panel
- Progressive Web App support for installable apps

Final Judging Notes

- **Demo the real flow.** Judges will place an order, accept it as a rider, and manage it as an admin, or sign up as a store owner, or take a mock test, depending on the problem.
- **UI is tested live** on a phone-sized screen and a desktop screen.
- **Deployed teams** are tested on their public link, and earn the extra deployment bonus.
- **Tie-breaker:** the team with the more polished UI and the more original feature wins.
- **Github:** Make repo as your teamname_Cypher and share the access to mystartupwave on the github

Closing line: "Anyone can build features. Great teams build products people enjoy using."